

Asset CIA Analysis (10 minutes)

Asset	Confidentiality (1-5)	Integrity (1-5)	Availability (1-5)	Justification <i>Why did you rate it this way?</i>
Pressure/flow readings	4	5	2	I think the integrity of the data has to be spot on because this is the baseline of possible accidents. The wrong data can either cause a shortage of water or create a very debilitating accident. Access to commands on the dashboard can also lead to a lot of funny events.
Gate control commands	5	5	3	This controls whether there would be water or not. Integrity of data is important, and also who can view this data is important. If the wrong person gets hold of this data (confidentiality), he or she can send wrong commands (integrity), and ultimately, a lot can happen. A short outage of this data may not cause a lot of damage.
Emergency shutoff capability	5	5	5	This is a safety-critical function. It must always be available in an emergency, and it must execute exactly as intended. If someone unauthorised gains access or the command is altered, it could either trigger unnecessary shutdowns or prevent shutdown during a real emergency, both of which are dangerous.
Dashboard login credentials	5	5	3	If someone who is not authorised has access to this, he or she can see all data that is being processed and also can understand the system and understand weak points in the system
Consumption/savings data	2	4	3	This data can reveal usage patterns and how operations go about, so some confidentiality is needed. Integrity is important so there can be accurate billing, plan the right way, and make the best changes to the system. Availability is useful for monitoring trends, but a temporary loss does not immediately endanger safety.

Leak detection alerts	2	5	5	The data itself is not very sensitive, but it must be accurate and delivered immediately. Availability is critical because delays defeat the purpose of leak detection.
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Attack Mapping

For each attack type you learned, identify **one specific scenario** at The Grand Marina:

Attack Type	Specific Scenario	Which Asset Is Targeted	CIA Impact
Eavesdropping	An attacker listens to unencrypted wireless traffic between sensors and the control dashboard on the marina network.	flow readings, consumption data	Confidentiality
Spoofing	An attacker impersonates a legitimate sensor and sends fake pressure readings to the system	Pressure readings	Integrity
Replay Attack	An attacker captures a valid “open gate” command and replays it later to release water at an unsafe time	Gate control	Integrity and availability (in a way)
Man-in-the-Middle	An attacker intercepts communication between the water meters and the dashboard and alters water consumption values before they reach the system, increasing or reducing the reported total cost.	Consumption/saving data (total water cost)	Integrity and confidentiality
Denial of Service	An attacker floods the control server with traffic, preventing real control commands or alerts from being processed.	Emergency shutoff capability, leak detection alerts	Availability
Unauthorized Access	An attacker gains access using stolen or weak dashboard login credentials	Dashboard login credentials, gate control commands	Confidentiality, Integrity, Availability

Priority Ranking

Based on your analysis, rank the six attacks from **most dangerous** to **least dangerous** for The Grand Marina:

1. Unauthorised Access. (**Most dangerous because it affects the availability, integrity and confidentiality of data. The hacker is present throughout the entire system. It can give full administrative access.**)
2. Man-in-the-middle. (**Second most dangerous because the ability to change data and reframe it in ways that fit one's wrong motives can lead to a lot of disruption in the system**)
3. Replay attack. (**Its impact is really close to that of MITM, just that you can really change data. Mimicking an action can cause catastrophic problems as well.**)
4. Denial of Service. (**I feel like this is the easiest to do, but also very easy to stop. Overloading a system can easily destroy a system that does not have the required security patches installed. Simple solutions like Blackhole routing or Rate limiting can easily prevent it.**)
5. Spoofing. (**Fake sensors or devices can inject false data, but this is often limited in scope compared to MITM or full unauthorised access.**)
6. Eavesdropping. (**Primarily impacts confidentiality. On its own, it cannot really change the system's behaviour, though it can enable more serious attacks later.**)